Question 1

Create an array A with elements `(12, 13, 14, 15, 16)` and display them.

```
A <- c(12, 13, 14, 15, 16)
A
```

Output

```
[1] 12 13 14 15 16
```

Question 2

Find the sum of all elements of `A`

```
sum(A)
```

Output

```
[1] 70
```

Question 3

Find the product of all elements of `A`

```
prod(A)
```

Output

```
[1] 524160
```

Question 4

Find the maximum and minimum element of `A`

```
max(A)
min(A)
```

Output

```
[1] 16[1] 12
```

Question 5

Find the range of array `A`

```
range(A)
```

Output

```
[1] 12 16
```

Question 6

Find the mean, variance and standard deviation and median of values of `A`

```
mean(A)
var(A)
sd(A)
```

Output

```
[1] 14[1] 2.5[1] 1.581139
```

Question 7

Sort the elements of A in both increasing and decreasing order and store them in `B` and `C`

```
B <- sort(A)
B

C <- sort(A, decreasing=TRUE)
C
```

Output

```
[1] 12 13 14 15 16[1] 16 15 14 13 12
```

Question 8

Create a matrix of 3x4 to have the set of natural numbers

```
mat <- matrix(seq(1, 12), nrow=3, ncol=4)
print(mat)
```

Output

```
     [,1] [,2] [,3] [,4]
[1,]    1    4    7   10
[2,]    2    5    8   11
[3,]    3    6    9   12
```

Question 9

Create MxN matrix by combining A, B and C row-wise (RW) and column-wise (CW).

```
mat2 <- matrix(A, B, C)
print(mat2)
```

Output

```
Warning message in matrix(A, B, C):
“data length [5] is not a sub-multiple or multiple of the number of rows [12]”
      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11] [,12] [,13]
 [1,]   12   14   16   13   15   12   14   16   13    15    12    14    16
 [2,]   13   15   12   14   16   13   15   12   14    16    13    15    12
 [3,]   14   16   13   15   12   14   16   13   15    12    14    16    13
 [4,]   15   12   14   16   13   15   12   14   16    13    15    12    14
 [5,]   16   13   15   12   14   16   13   15   12    14    16    13    15
 [6,]   12   14   16   13   15   12   14   16   13    15    12    14    16
 [7,]   13   15   12   14   16   13   15   12   14    16    13    15    12
 [8,]   14   16   13   15   12   14   16   13   15    12    14    16    13
 [9,]   15   12   14   16   13   15   12   14   16    13    15    12    14
[10,]   16   13   15   12   14   16   13   15   12    14    16    13    15
[11,]   12   14   16   13   15   12   14   16   13    15    12    14    16
```

```
[12,]   13   15   12   14   16   13   15   12   14    16    13    15    12
      [,14] [,15] [,16]
 [1,]    13    15    12
 [2,]    14    16    13
 [3,]    15    12    14
 [4,]    16    13    15
 [5,]    12    14    16
 [6,]    13    15    12
 [7,]    14    16    13
 [8,]    15    12    14
 [9,]    16    13    15
[10,]    12    14    16
[11,]    13    15    12
[12,]    14    16    13
```

Question 10

: Find the 10 and 3 row elements of RW

Question 11

: Find 1 and 4 columns of CW

Question 12

: Using both RW and CW find sub-matrices having elements `[2, 3]` and `[2, 4]`